

Niraiva OnCoTrack

MVP Specification for Investors, Clinicians, and Partners

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Product: A module embedded in the existing Niraiva healthcare portal

Status: MVP definition for investment, design-partner, and implementation planning

Executive Summary

Niraiva OnCoTrack is a clinician-supervised cancer-signal intelligence and follow-up module inside the Niraiva portal. It turns fragmented symptoms, screening history, routine lab results, specialty biomarkers, imaging, and pathology records into a longitudinal view—then makes sure each review-worthy signal has a named owner, next step, and documented outcome.

The MVP does **not** claim to detect cancer, provide a cancer probability score, replace screening, or make diagnostic decisions. Its initial value is more practical and more defensible: prevent clinically important signals and overdue screenings from being lost between the lab result, patient, ordering clinician, specialist, and follow-up visit.

This is a workflow and intelligence product, not a laboratory or diagnostic assay. Niraiva integrates and explains information produced elsewhere, preserves the original source data, supports clinician judgment, and closes the loop on care.

1. The Problem

Cancer care has a critical operational gap before diagnosis: abnormal or incomplete signals often exist in separate systems, but no one continuously assembles the history, recognizes what requires review, assigns ownership, and proves that follow-up happened.

A patient may have:

- A PSA result ordered by primary care.
- A CBC or metabolic panel from urgent care.

- A CA-125 ordered by a specialist.
- A missed mammogram, colorectal screen, cervical screen, or lung-screening eligibility signal.
- New symptoms recorded in one visit and imaging or pathology in another.
- A privately ordered advanced blood-based cancer test in a PDF report.

Today, these data may be isolated by lab, facility, clinician, time period, or patient portal. Most EHR portals show individual raw values and reference ranges. They are not designed to create a neutral, multi-source longitudinal signal history and a closed-loop accountability workflow across the patient's care journey.

The consequence is not merely inconvenience. It creates avoidable diagnostic delay, patient anxiety, repetitive outreach, missed screening opportunities, incomplete referrals, and safety/liability exposure from unclosed abnormal-result follow-up.

2. The MVP Solution

OnCoTrack adds five coordinated capabilities to the existing Niraiva charting portal:

MVP capability	What it does	Why it matters
Longitudinal signal timeline	Brings relevant lab results, symptoms, screening history, imaging, pathology, and follow-up events into one chronological chart view	Clinicians can see patterns and prior workup without hunting across systems
Explainable trend review	Plots comparable results over time, keeps source labs and reference ranges visible, and identifies review-worthy changes	A trend can matter even when a single result is not clearly abnormal; the clinician sees the underlying evidence
Screening-gap engine	Identifies potentially due or overdue guideline-based screening independent of laboratory findings	Normal blood work must never be mistaken for being up to date on recommended screening
Closed-loop navigation	Assigns owner, next step, due date, escalation, and final disposition for each review-worthy item	Prevents "alert and abandon" and creates measurable accountability
Clinician-supervised AI assistance	Extracts verified data from documents, drafts timelines and referral packets, and explains approved rules with citations	Reduces administrative burden without allowing AI to diagnose, order, or close care cases autonomously

The core user journey

1. **Ingest:** Niraiva receives lab data, chart data, patient-reported information, imaging/pathology documents, and approved external reports.

2. **Normalize:** The module preserves the original result and maps it into a common, auditable data model.
3. **Trend and assess:** It compares only compatible results, displays patterns over time, checks screening status, and identifies items requiring clinician review.
4. **Review:** A clinician sees the exact source results, timing, method differences, known non-cancer explanations, missing context, and applicable local workflow.
5. **Route:** The clinician documents a plan, assigns responsibility, and may authorize referral, repeat testing, screening outreach, monitoring, or no current action.
6. **Close:** Niraiva tracks work to documented disposition—such as completed screening, specialist assessment, benign explanation, transfer of care, patient decline, or confirmed diagnosis.

The MVP's success is not "how many cancers it predicts." It is whether Niraiva reliably closes the gap between a signal and a documented clinical decision.

3. What Ships First

In scope

A. Tier A laboratory intelligence

The MVP ingests and trends established clinical markers and routine contextual tests, including:

- PSA, CA-125, CEA, AFP, CA 19-9, CA 15-3/CA 27.29, beta-hCG, calcitonin, thyroglobulin, HE4, NSE/ProGRP, LDH, beta-2 microglobulin, serum free light chains, SPEP/immunofixation, chromogranin A, SCC antigen, and CYFRA 21-1.
- CBC with differential: hemoglobin, RBC/WBC/platelet counts, red-cell indices, and differential components.
- CMP/chemistry: liver enzymes, alkaline phosphatase, bilirubin, albumin, calcium, renal function, and available components.
- Contextual inflammation/nutrition indicators such as CRP, ESR, and ferritin.

These results are used as **contextual clinical signals**, not population cancer screening tests. Many are nonspecific and may change because of infection, inflammation, pregnancy, renal or liver disease, benign conditions, medications, smoking, recent procedures, or laboratory-method differences.

B. Structured symptom and sign intake

Clinicians can capture discrete, queryable patient observations instead of relying only on free text:

- Systemic signs: unexplained weight loss, fatigue, fever, night sweats, persistent or progressive pain.
- Surface findings: new or changing lumps, mole change, non-healing sore, skin change, jaundice.
- Functional symptoms: unusual bleeding, blood in stool/urine/sputum, persistent cough, hoarseness, swallowing difficulty, bowel or urinary changes, indigestion, eating difficulty.
- Breast-specific symptoms: lump, thickening, skin or shape change, nipple changes/discharge, breast pain, location, size, duration, and progression.

C. Screening-gap tracking

The MVP identifies potential gaps in breast, colorectal, cervical, lung, and prostate-related screening workflows using a versioned, institution-approved guideline package. The engine distinguishes:

- Due.
- Overdue.
- Not due.
- Not eligible.
- Unknown because required history is missing.
- Excluded because a documented clinical reason applies.
- Needs clinician review.

As examples of the type of guidance a tenant may configure, current USPSTF recommendations include biennial mammography for women ages 40–74, colorectal screening for adults ages 50–75, cervical screening options for women ages 21–65, and annual low-dose CT for eligible adults ages 50–80 with qualifying smoking history. Niraiva will use locally approved, versioned rule packages rather than hard-coding a static public guideline. [web:19][web:21]

D. Imaging, pathology, and oncology-charting support

The MVP allows structured capture and source-document linking for:

- Imaging method, lesion location/size/shape/margins, nodes, and reported distant findings.
- Biopsy and pathology results, specimen type, histology, tumor classification, and source report.

- Breast-cancer receptor fields where documented: HER2, ER, PR, IHC/FISH method, result, percentages, ratio, and ERBB2 notes.
- TNM elements, overall stage if clinician-entered or sourced from an approved report, nodal involvement, and metastasis information.

This feature supports continuity once evaluation is underway. It does not independently interpret imaging, pathology, receptor status, or stage disease.

E. Closed-loop care navigation

Every review-worthy item becomes a tracked **Signal Follow-up Case** with:

- A named owner and team.
- Priority and due date.
- Specific evidence that initiated the case.
- A clinician review and documented disposition.
- Follow-up tasks, referral status, outreach history, and escalation if overdue.
- Closure evidence and a complete audit trail.

F. Clinician portal and patient-facing experience

Clinicians receive a longitudinal evidence view, a triage worklist, a case action panel, and a referral-packet draft workflow. Patients see only clinician-released information, approved education, follow-up tasks, appointments, consent choices, and secure communications. Patients do not see unreviewed alerts, AI drafts, or a cancer-risk score.

Explicitly out of scope

Excluded from MVP	Why
Autonomous cancer probability score	Scientifically fragile without validation and creates substantially greater regulatory and safety exposure
Autonomous diagnosis, staging, treatment recommendation, or biopsy recommendation	Must remain a clinician decision supported by source evidence and approved protocols
Direct-to-patient “cancer detected” alerts	Creates false reassurance and unnecessary alarm; clinician review is required first

Proprietary liquid-biopsy / MCED assay	This is a laboratory science business, not Niraiva's initial software wedge
Independent tissue-of-origin prediction	Belongs to the underlying validated assay/report, not the portal
Routine hereditary/genetic risk module	Requires separate consent, counseling pathways, access controls, and legal review; deferred to Phase 2
Research-only biomarkers	Clinical utility is insufficient for MVP decision support
Automatic ordering/referrals/messages	Uses existing Niraiva authorization workflow; no AI or rules engine may act without authorized user action

4. How It Works Safely

Three separate lanes—not one alarming score

OnCoTrack keeps three types of work separate so a missed screening is never confused with a concerning trend, and a nonspecific result is never represented as cancer risk.

Lane	What it means	Example	Required outcome
Preventive screening gap	A patient may be due or overdue for guideline-based screening	Prior screening date absent or interval has elapsed	Screening plan, exception, or missing-data resolution documented
Clinical-concern trend	Comparable data show a pattern requiring a clinician's contextual review	Repeated compatible values changing over time, or a clinically approved multi-result pattern	Clinician review and plan documented
Marker-specific follow-up	A source result or report requires tracking under local policy or ordering workflow	An abnormal report needing repeat testing, specialist review, or documented handoff	Disposition, handoff, or follow-up completion documented

Explainability is mandatory

For every alert or workflow item, the clinician can see:

- The source results, dates, units, reference ranges, laboratory/method, and original report.
- The patient-specific trend graph and calculation when comparison is valid.

- Whether data are comparable, limited, incomplete, corrected, or incompatible.
- Known non-cancer causes that may affect the finding.
- Missing patient context that could materially change interpretation.
- The exact rule, evidence-content version, and guideline/workflow version used.
- The accountable owner, due date, and next action.

This approach aligns with FDA’s clinical decision-support framework, which emphasizes that eligible non-device CDS should enable a healthcare professional to independently review the basis for a recommendation rather than relying primarily on opaque software output. [web:22][web:24]

Human review is non-negotiable

No workflow can:

- Send an unreviewed cancer-related finding to a patient.
- Close a case without a documented disposition and evidence of resolution, handoff, or patient decision.
- Convert an AI extraction into a clinical fact without verification.
- Generate a diagnosis, cancer probability, tissue-of-origin conclusion, staging conclusion, treatment plan, or mandatory biopsy directive.
- Treat missing information as negative, normal, low risk, or resolved.

5. Decision Workflow and Risk Models

Clinical workflow engine: build the infrastructure, not an unvalidated protocol

Niraiva has a draft early-malignancy workflow matrix that describes a possible sequence from intake and risk stratification through screening, diagnostic workup, pathology, staging, and treatment handoff. The source document itself is informational and not a validated clinical guideline.

Therefore, the MVP builds a **versioned clinical workflow engine in shadow mode**:

- It can represent nodes such as intake, high-risk review, standard screening, abnormal imaging, surveillance, pathology, benign finding, confirmed malignancy, staging, and oncology-care handoff.

- Every rule is versioned, tied to evidence and approval, and disabled by default.
- Before a rule can affect live workflow, a designated medical director or clinical governance board must approve it.
- In shadow mode, the platform shows what the workflow would have suggested on de-identified or test data without changing the patient record or creating patient-facing output.
- Clinicians may always override an approved workflow with a documented rationale.

The architecture supports future implementation of high-risk cohort workflows, specialty review, and time-bound navigation targets. It deliberately does not hard-code universal claims such as “all patients must have biopsy within seven days.” Any such action or SLA must be approved and configured by the deploying organization.

Probability-table infrastructure: not a patient-facing MVP feature

The supplied lung-cancer example contains probabilities based on gender and smoking status, but it has an incomplete value for male nonsmokers and no clinical-validation package. The MVP will support the **data-governance infrastructure** for future validated risk tables, but will not expose a generic patient cancer-risk score.

Required controls:

- Every table must identify source, population, inputs, version, effective date, validation evidence, clinical owner, and approval status.
- Unknown reference values return “insufficient reference data,” never zero or a silent default.
- Values cannot be used in patient-facing communication or workflow automation until clinically approved.
- Adding variables such as age, pack-years, quit date, family history, or comorbidity requires model validation and governance—not simply adding fields to a table.

6. Why This Is Investable

A clear buyer problem

Health systems, value-based care organizations, ACOs, and self-insured employers already bear the cost of:

- Missed preventive screenings.
- Unresolved abnormal results and referrals.
- Duplicate testing and fragmented records.
- Navigation burden on nurses, PCPs, specialists, and care managers.
- Quality-measure gaps and diagnostic-delay risk.

OnCoTrack sells workflow reliability and measurable follow-up completion—not speculative cancer prediction.

Differentiated market position

Category	What they primarily provide	Niraiva OnCoTrack's role
Liquid-biopsy / MCED companies	Proprietary blood assay and test report	Vendor-neutral routing, longitudinal context, and follow-up tracking across all sources
Consumer biomarker apps	General wellness dashboards and panels	Clinically governed, chart-integrated review and accountable care navigation
EHR portals	Raw result display and basic messaging	Multi-source trend intelligence, evidence explanation, cross-clinician follow-up ownership, and closure measurement
Oncology navigation vendors	Support after a cancer diagnosis	Pre-diagnosis signal and screening follow-up, then handoff to active oncology care when appropriate

Compounding data advantage

The product becomes more valuable over time through three assets:

1. A governed, versioned biomarker and screening knowledge base.
2. A longitudinal, multi-source clinical timeline that retains comparability and provenance.
3. Closed-loop outcome data: what signal was observed, what action was chosen, whether it was completed, and what disposition occurred.

Any future research or data-use program must be optional, appropriately consented, legally reviewed, and de-identified; it is not an MVP revenue assumption.

Business model

Revenue stream	Buyer	MVP status
Health-system / ACO / employer license	Health systems, value-based organizations, self-insured employers	Primary MVP commercial model
Practice subscription	Primary care, women's health, oncology-adjacent and concierge practices	MVP-capable secondary channel
Implementation and integration services	Design partners	Limited, structured onboarding revenue
Consumer premium navigation	Consumers, only through clinician-anchored experience	Potential later phase; not the core launch motion
Lab/MCED routing partnerships	Labs and assay companies	Future expansion after workflow traction
Opt-in de-identified research data	Research/life-science partners	Long-term, governance-dependent; not Year 1 plan

7. MVP Product Architecture

OnCoTrack is an additive module within the existing Niraiva portal. It reuses existing patient identity, provider identity, authentication, role-based access, encounters, chart UI, notes, document storage, tasks, notifications, messaging, audit logging, and APIs wherever those services already exist.

New services and data domains

Component	Purpose
Lab and biomarker ingestion service	Receives HL7/FHIR/API/document data; preserves raw source and normalized representation
Trend service	Builds patient-specific longitudinal visualizations only for comparable data
Symptom and sign module	Stores structured clinical observations with duration, progression, location, and source
Screening-rule engine	Evaluates approved, versioned screening criteria and identifies missing eligibility data
Imaging/pathology module	Stores structured, source-linked findings, receptor status, and TNM elements when applicable
Clinical workflow engine	Represents approved care pathways and runs in shadow mode until governance activates specific rules

Signal Follow-up Case service	Tracks ownership, tasks, referral status, escalations, dispositions, and closure
Evidence knowledge base	Holds biomarker associations, confounders, LOINC mappings, permitted uses, rule versions, and review dates
AI assist service	Supports source-cited extraction, drafting, and summarization with human verification
Governance console	Controls approval, versioning, audit, effective dates, rollback, and tenant-specific configuration

Interoperability requirements

- Use Niraiva’s existing APIs, identity system, and chart events first.
- Support HL7 v2 laboratory-result messages and FHIR Observation, DiagnosticReport, DocumentReference, Task, ServiceRequest, CarePlan, Consent, Provenance, and related resources as appropriate.
- Preserve original lab name, specimen, method, unit, range, collection date, result status, and report attachment.
- Quarantine unmapped, preliminary, duplicated, cancelled, amended, or unverified OCR results until the system can safely classify them.
- Use versioned mappings and terminology. USCDI v3 is the current baseline adopted under ONC’s HTI-1 certification framework, which supports a standards-based integration strategy. [web:18]

8. AI Features and Constraints

What AI can do

AI capability	MVP behavior	Safety control
Document extraction	Extract candidate labs, dates, symptoms, imaging descriptors, pathology fields, and follow-up recommendations from PDFs	Human verifies source-linked values before clinical use
Timeline summary	Draft a concise chronology of results, symptoms, workup, and open tasks	Every statement links to chart evidence; clinician must review before filing
Referral packet drafting	Assemble verified information into a specialist-ready packet	Clinician reviews, edits, and authorizes transmission

Rule explanation	Explain why an approved rule created a review item	Shows source values, rule version, limitations, confounders, and missing inputs
Operational queue support	Sort cases by due date, case age, SLA risk, and documented workflow priority	Cannot infer diagnosis or independently alter clinical urgency

What AI cannot do

- Diagnose or exclude cancer.
- Assign a cancer likelihood or probability score.
- Generate a tissue-of-origin conclusion.
- Interpret a molecular test beyond displaying the authorized source report and approved explanatory content.
- Change case severity, close a case, or alter a clinician disposition.
- Order tests, place referrals, prescribe, or send patient messages without explicit authorized user action.
- Learn from patient PHI for training or fine-tuning without explicit legal, contractual, privacy, and consent authorization.

9. MVP Success Measures

Primary metric

At least 90% of review-worthy clinical-concern and marker-specific follow-up cases receive a documented clinician disposition within the pilot's approved service-level target. A 14-day target can be tested during pilot design, but each tenant's final SLA must be clinically governed.

Additional pilot measures

- Time from eligible source result to clinician review.
- Time from clinician review to documented action plan.
- Percentage of due/overdue screenings with documented plan, completion, or justified exception.

- Percentage of cases closed with valid evidence.
- Overdue case count, escalation rate, and reopen rate.
- Lab mapping accuracy, OCR verification accuracy, duplicate detection, and source-data latency.
- Clinician trust, usability, alert relevance, and override/suppression reasons.
- Patient communication comprehension and navigation satisfaction, where measured.
- PHI security incidents: target zero.

Claims the MVP will not make

Until supported by prospective clinical evidence and appropriate review, Niraiva will not claim:

- Cancer sensitivity, specificity, PPV, NPV, or detection accuracy.
- Mortality reduction or stage shift.
- Reduced biopsy rates or diagnostic yield.
- Cancer prevention or diagnosis.
- Superiority to clinician judgment or standard-of-care evaluation.

10. Development Plan

Phase	Deliverable
Foundation	Architecture review, intended-use statement, safety plan, governance process, role model, consent/audit design
Data foundation	FHIR/HL7 ingestion, source provenance, Tier A knowledge base, mapping and OCR verification workflow
Clinical experience	OnCoTrack chart tab, longitudinal timeline, trend graphs, symptom intake, worklist, screening view
Closed loop	Follow-up cases, tasks, ownership, dispositions, escalations, referral-packet workflow
Governed workflow and AI	Shadow-mode pathway engine, governance console, source-cited document/timeline/referral AI assistance
Pilot hardening	Validation, security testing, usability testing, interface reconciliation, training, dashboards, go-live readiness

First pilot profile

The ideal first design partner is a health system, primary-care network, women's-health network, or oncology-adjacent practice with:

- An identifiable abnormal-results follow-up problem.
- Sufficient lab and screening data access.
- A clinical champion
- Nurse navigator or care-management capacity.
- Willingness to run a governed workflow pilot and measure closure rates.

11. Key Risks and Mitigations

Risk	Mitigation built into MVP
Regulatory classification risk	No autonomous diagnosis or risk score; transparent source data; clinician reviews basis and makes decisions; claims controlled by governance
False reassurance or anxiety	No patient-facing unreviewed alerts; explicit limitations and confounders; separate lanes instead of one "cancer score"
Alert fatigue	Comparability checks, deduplication, required evidence, configurable thresholds, suppression with expiration, and relevance tracking
Poor lab-data quality	Preserve original data; quarantine unclear/missing mappings; do not compare incompatible assays; visible data-quality states
AI hallucination or unsafe extraction	Source citations, schema validation, clinician verification, prohibited output controls, and deterministic non-AI fallback
Long enterprise sales cycles	Begin with 1–2 design partners and a tightly scoped measurable pilot
Unvalidated workflow content	Versioned rules run in shadow mode until clinical governance approves them; no hard-coded draft protocol actions
Sensitive-data exposure	Niraiva identity/RBAC, minimum necessary access, encryption, consent, audit logs, BAAs, and special controls for pathology/staging/genetic data
Assay-company competition	Niraiva is neutral infrastructure that can work across labs, assays, EHR sources, and care settings

12. Investment Ask Framework

Intended seed-round use of funds

Funding supports:

- Engineering: ingestion pipeline, longitudinal trend engine, chart-native clinician experience, care-navigation workflow, integration APIs, and security hardening.
- Clinical and regulatory: medical director, clinical advisory board, evidence knowledge-base governance, legal/regulatory counsel, and pre-submission strategy if needed.
- Pilot execution: 1–2 design partners, implementation, integration, training, support, and outcome measurement.
- Trust infrastructure: HIPAA/security program, auditability, consent management, quality systems, and AI safety evaluation.

Funded milestones

1. MVP live inside Niraiva with one or two governed design partners.
2. Tier A lab and screening data ingested into an explainable longitudinal timeline.
3. Closed-loop Signal Follow-up Cases demonstrating the target clinician-disposition completion rate.
4. Screening-gap workflow operating with approved local guideline packages.
5. Shadow-mode clinical workflow engine validated on test/de-identified data.
6. Security, audit, consent, and clinical-governance controls operational.
7. Evidence package sufficient to pursue additional health-system, payer, employer, lab, or strategic distribution discussions.

13. The Simple Explanation

For a patient: **Niraiva helps your care team make sure important test results, symptoms, and overdue screenings do not get lost—and that there is a documented next step.**

For a doctor: **Niraiva gives you source-linked trends, screening context, and a closed-loop work queue, while you retain clinical judgment and control.**

For a health system: **Niraiva is an abnormal-result and screening-follow-up operating layer that reduces fragmentation, documents accountability, and creates measurable care-navigation performance.**

For an investor: **Niraiva owns the neutral software layer between an emerging flood of cancer-related signals and the real-world clinical workflows required to resolve them.**

Appendix: Clinical and Product Boundaries

The MVP includes structured symptoms, routine lab panels, established tumor-marker contexts, screening gaps, imaging/pathology record support, and clinician-directed navigation. The supplied cancer-related dataset is the source for the initial symptom, breast-imaging, pathology, receptor, TNM, and observation-metadata fields. [file:1]

The supplied early-stage malignancy node matrix is incorporated only as a configurable workflow-template concept. Its nodes, transition logic, clinical thresholds, workup instructions, and stated timing rules require medical-director and clinical-advisory-board review before any live behavior. [file:2]

The product must not present itself as medical advice, a clinical diagnosis, a replacement for recommended screening, or a substitute for qualified clinician evaluation. Regulatory classification, claims, workflow rules, evidence content, and patient communication must be reviewed by qualified clinical, legal, privacy, and regulatory leaders before deployment.